

Stochastic Approximation Analysis of NEP Student Evolution

Version: 2026-06-28 | **Status:** Theoretical framework | **Scope:**
Analysis of the coupled NEP student / gatekeeper evolution using stochastic approximation and two-timescale ODE methods
Notation note: $\mathcal{S}$ denotes the state space (existing SCX theory). S_t denotes the gatekeeper scoring function at time t . f_{θ_t} denotes the NEP student with parameters θ_t .

Table of Contents

1. Introduction: The Coupled Learning System
 2. Robbins-Monro Formulation of NEP Student Update
 3. Learning Rate Conditions
 4. Gradient Properties and Boundedness
 5. Convergence to Stationary Point
 6. Distribution Shift Analysis
 7. Two-Timescale Analysis
 8. ODE Method
 9. Connection to State-Conditioned Risk
 10. Full Proof Sketch: Convergence Under Robbins-Monro Conditions
 11. What Breaks When Distribution Shifts Too Rapidly
 12. Summary of Proven vs. Conjectured Claims
 13. References
-

1. Introduction: The Coupled Learning System

The self-evolution loop creates a **coupled dynamical system** between the gatekeeper S_t and the NEP student f_{θ_t} :

$$\begin{aligned} S_{t+1} &= \text{UpdateGatekeeper}(S_t, \mathcal{M}_t, f_{\theta_t}) \\ \theta_{t+1} &= \theta_t - \alpha_t \nabla_{\theta} \mathcal{L}(f_{\theta_t}, \mathcal{M}_t) \end{aligned}$$

This document analyzes the student evolution via **stochastic approximation** (Robbins & Monro, 1951), treating the gatekeeper update as a slow drift in the data distribution.

The key challenge: The data distribution for the student is not stationary. As S_t evolves, the composition of $\mathcal{M}_t$ changes, which shifts the effective training distribution for f_{θ_t} . Standard stochastic approximation theory assumes a stationary target; we must extend it to handle this drift.

2. Robbins-Monro Formulation of NEP Student Update

Definition 5.1 (Stochastic Gradient Update). The NEP student parameters $\theta_t \in \mathbb{R}^{d_{\theta}}$ evolve according to:

$$\theta_{t+1} = \theta_t - \alpha_t \cdot g_t(\theta_t)$$

where:

- $\alpha_t > 0$ is the learning rate (step size) at time t

- $g_t(\theta_t) = \nabla_{\theta} \ell(f_{\theta_t}(x_t), y_t)$ is the stochastic gradient at a single sample $(x_t, y_t) \sim P_t$
- P_t is the data distribution at time t , induced by the current gatekeeper S_t

Assumption SE-A5 (Robbins-Monro Noise Model). The stochastic gradient $g_t(\theta_t)$ satisfies:

$$g_t(\theta_t) = \nabla L_t(\theta_t) + \xi_t$$

where:

- $L_t(\theta) = \mathbb{E}_{(x,y) \sim P_t} [\ell(f_{\theta}(x), y)]$ is the expected loss under the current data distribution
- $\nabla L_t(\theta_t)$ is the true gradient of the expected loss at θ_t
- ξ_t is zero-mean noise: $\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$
- $\text{Var}(\xi_t \mid \mathcal{F}_{t-1}) \leq \sigma_{\xi}^2 < \infty$ (bounded variance)

Remark. The expected loss L_t is time-dependent because P_t depends on S_t , which evolves. This is the central complication vs. standard SA.

3. Learning Rate Conditions

Definition 5.2 (Robbins-Monro Learning Rate). The learning rate α_t satisfies the standard Robbins-Monro conditions:

$$\sum_{t=1}^{\infty} \alpha_t = \infty, \quad \sum_{t=1}^{\infty} \alpha_t^2 < \infty$$

Examples. Common choices satisfying these conditions:

1. **Polynomial decay:** $\alpha_t = \alpha_0 \cdot t^{-\gamma}$, where $\gamma \in (1/2, 1]$. The most common choice is $\gamma = 1$ (harmonic series).
2. **Logarithmic decay:** $\alpha_t = \alpha_0 \cdot (\log(t + 1))^{-1}$.
3. **Constant learning rate:** $\alpha_t = \alpha$ (for tracking non-stationary distributions). This does NOT satisfy $\sum \alpha_t^2 < \infty$ and leads to different convergence behavior (see Section 6).

Assumption SE-A6 (Decaying Learning Rate). For asymptotic convergence analysis, we assume α_t satisfies the RM conditions. For the tracking/constant-rate case, we relax this and analyze the asymptotic bias.

Proposition 5.1 (Constant vs. Decaying Learning Rate Trade-off).

Learning Rate	Convergence	Tracking Ability
Decaying ($\sum \alpha_t^2 < \infty$)	Almost sure to stationary point	Poor: cannot track drift
Constant ($\alpha_t = \alpha$)	Bounded in distribution around optimum	Good: adapts to drift

Proof sketch. For decaying rates, the RM convergence theorem (Theorem 5.1 below) gives almost sure convergence but requires vanishing noise, which precludes tracking a moving target. For constant rates, the iterate converges in distribution to a stationary process whose expected distance to the optimum is $O(\alpha)$ (Kushner & Yin, 2003, Chapter 5).

4. Gradient Properties and Boundedness

Assumption SE-A7 (Gradient Lipschitz Continuity). The loss function $\ell(f_\theta(x), y)$ is once-differentiable in θ with Lipschitz continuous gradient:

$$\|\nabla_\theta \ell(f_{\theta_1}(x), y) - \nabla_\theta \ell(f_{\theta_2}(x), y)\| \leq L_g \|\theta_1 - \theta_2\|, \quad \forall \theta_1, \theta_2$$

for some $L_g < \infty$, uniformly over (x, y) in the support of the data distribution.

Assumption SE-A8 (Gradient Boundedness). The stochastic gradient is uniformly bounded:

$$\|g_t(\theta)\| \leq G < \infty, \quad \forall \theta \in \mathbb{R}^{d_\theta}, \forall t$$

Proposition 5.2 (Consequences of Lipschitz Gradients). Under SE-A7:

1. **Expected loss smoothness:** $L_t(\theta)$ has L_g -Lipschitz gradient for all t :

$$\|\nabla L_t(\theta_1) - \nabla L_t(\theta_2)\| \leq L_g \|\theta_1 - \theta_2\|$$

2. **Descent inequality:**

$$L_t(\theta_{t+1}) \leq L_t(\theta_t) + \langle \nabla L_t(\theta_t), \theta_{t+1} - \theta_t \rangle + \frac{L_g}{2} \|\theta_{t+1} - \theta_t\|^2$$

3. **Expected one-step improvement:**

$$\mathbb{E}[L_t(\theta_{t+1}) \mid \mathcal{F}_{t-1}] \leq L_t(\theta_t) - \alpha_t \|\nabla L_t(\theta_t)\|^2 + \frac{L_g \alpha_t^2}{2} \mathbb{E}[\|g_t(\theta_t)\|^2 \mid \mathcal{F}_{t-1}]$$

5. Convergence to Stationary Point

Theorem 5.1 (Convergence of Robbins-Monro for NEP Student).

Let assumptions SE-A5 (noise model), SE-A6 (decaying learning rate), SE-A7 (Lipschitz gradient), and SE-A8 (bounded gradient) hold. Additionally, assume that the data distribution P_t converges to a limiting distribution P_∞ as $t \rightarrow \infty$ (or equivalently, $S_t \rightarrow S_\infty$). Define the limiting expected loss:

$$\bar{L}(\theta) = \mathbb{E}_{(x,y) \sim P_\infty} [\ell(f_\theta(x), y)]$$

Then:

(a) **Gradient vanishing.** $\|\nabla L_t(\theta_t)\| \rightarrow 0$ almost surely as $t \rightarrow \infty$.

(b) **Stationary point.** Any limit point θ^* of the sequence $\{\theta_t\}$ satisfies:

$$\nabla \bar{L}(\theta^*) = 0$$

i.e., θ^* is a stationary point of the limiting expected loss.

(c) **Rate under strong convexity.** If $\bar{L}(\theta)$ is μ -strongly convex, then:

$$\|\theta_t - \theta^*\| = O\left(\frac{1}{\sqrt{t}}\right) \quad \text{in probability}$$

Proof sketch. See Section 10 for the full proof sketch. The key steps are: 1. Show that $\sum \alpha_t \langle \nabla L_t(\theta_t), \mathbb{E}[g_t(\theta_t) \mid \mathcal{F}_{t-1}] \rangle$ converges using the supermartingale convergence theorem. 2. Relate this to $\sum \alpha_t \|\nabla L_t(\theta_t)\|^2$ using the noise decomposition. 3. Conclude $\|\nabla L_t(\theta_t)\| \rightarrow 0$ from $\sum \alpha_t = \infty$. 4. Show convergence of L_t to $\bar{L}$ from $P_t \rightarrow P_\infty$.

Corollary 5.1 (Local Minima). Under the additional assumption that $\bar{L}$ has isolated local minima, θ_t converges to a local minimum θ^* almost surely, provided the initialization is in the basin of attraction.

6. Distribution Shift Analysis

Definition 5.3 (Distribution Shift Operator). The distribution P_t evolves according to the gatekeeper dynamics:

$$P_t(x, y) = \frac{S_t(x, y) \cdot P_0(x, y)}{\mathbb{E}_{P_0}[S_t]}$$

where P_0 is the base data distribution and S_t is the gatekeeper at time t . This is the acceptance-biased distribution.

Theorem 5.2 (Distribution Shift Bound). Let $\|S_t - S_\infty\|_\infty \leq \varepsilon_t$ where $\varepsilon_t \rightarrow 0$ (from the martingale convergence of S_t). Then the total variation distance between P_t and P_∞ satisfies:

$$\text{TV}(P_t, P_\infty) \leq \frac{2\varepsilon_t}{\mathbb{E}_{P_0}[S_\infty] - \varepsilon_t}$$

Proof. By definition:

$$P_t(x, y) = \frac{S_t(x, y)P_0(x, y)}{Z_t}, \quad P_\infty(x, y) = \frac{S_\infty(x, y)P_0(x, y)}{Z_\infty}$$

where $Z_t = \mathbb{E}_{P_0}[S_t]$, $Z_\infty = \mathbb{E}_{P_0}[S_\infty]$. Then:

$$\begin{aligned}
 |P_t(x, y) - P_\infty(x, y)| &\leq \left| \frac{S_t}{Z_t} - \frac{S_\infty}{Z_\infty} \right| P_0(x, y) \\
 &\leq \left(\frac{|S_t - S_\infty|}{Z_t} + \frac{S_\infty |Z_t - Z_\infty|}{Z_t Z_\infty} \right) P_0(x, y) \\
 &\leq \frac{2\varepsilon_t}{Z_\infty - \varepsilon_t} P_0(x, y)
 \end{aligned}$$

Integrating over (x, y) gives the TV bound.

Corollary 5.2 (Stationarity After Convergence). Since $S_t \rightarrow S_\infty$ almost surely (Theorem 4.4), we have $P_t \rightarrow P_\infty$ in total variation almost surely.

Impact on Student Training. The distribution shift means that earlier student updates were computed under P_i for $i < t$, while the convergence analysis is relative to P_∞ . The discrepancy is controlled by Theorem 5.2.

Proposition 5.3 (Cumulative Shift Error). The total error due to distribution shift up to time t is bounded by:

$$\sum_{i=1}^t \alpha_i \cdot \text{TV}(P_i, P_\infty) \leq \sum_{i=1}^t \frac{2\alpha_i \varepsilon_i}{\bar{Z} - \varepsilon_i}$$

where $\bar{Z} = \mathbb{E}_{P_0}[S_\infty] > 0$. If $\varepsilon_t = o(\alpha_t)$, the cumulative shift error is $o(1)$.

Proof. Direct application of Theorem 5.2 and summation.

7. Two-Timescale Analysis

Definition 5.4 (Two-Timescale Structure). The self-evolution system has a natural two-timescale separation:

Component	Update interval	Speed	Role
Gatekeeper S_t	$1/\nu_S$ updates per data batch	Slow (macro)	Defines the data distribution
NEP student θ_t	Every sample	Fast (micro)	Adapts to current distribution

Formally, define τ_S and τ_θ as the characteristic timescales:

$$\tau_S = O(1/\nu_S), \quad \tau_\theta = O(1/\alpha_t)$$

where ν_S is the gatekeeper update frequency (number of gatekeeper updates per student step). The two-timescale separation holds when:

$$\nu_S \ll \alpha_t \quad \text{or equivalently} \quad \tau_S \gg \tau_\theta$$

Theorem 5.3 (Two-Timescale Convergence). Under the two-timescale condition $\alpha_t \gg \nu_S$ (student learns much faster than gatekeeper updates), the coupled system can be analyzed as follows:

(a) Fast timescale (θ quasi-stationary given S). For fixed S , the student update converges to the minimizer of the expected loss under P_S :

$$\theta_t \rightarrow \theta^*(S) := \arg \min_{\theta} \mathbb{E}_{(x,y) \sim P_S} [\ell(f_{\theta}(x), y)]$$

(b) Slow timescale (S evolves with θ tracking). On the slow timescale, the gatekeeper dynamics are:

$$S_{t+1} \approx S_t + \nu_S \cdot \Delta(S_t, \theta^*(S_t))$$

where Δ is the gatekeeper update function evaluated at the quasi-stationary student.

(c) Reduced system. The coupled system reduces to a single-timescale system on S alone:

$$S_{t+1} = S_t + \nu_S \cdot \Delta(S_t, \theta^*(S_t))$$

with the student implicitly at $\theta^*(S_t)$.

Proof sketch. Formal application of singular perturbation theory (Khalil, 2002; Borkar, 2008). The fast subsystem (student) has a unique globally attracting fixed point $\theta^*(S)$ for each S . By Tikhonov's theorem, the coupled system converges to the slow manifold $\{\theta = \theta^*(S)\}$ as $\alpha_t/\nu_S \rightarrow \infty$, and the reduced system on the slow manifold captures the asymptotic behavior.

Corollary 5.3 (Separation Condition). The two-timescale approximation is valid when:

$$\frac{\alpha_t}{\nu_S} \gg \frac{1}{\mu}$$

where μ is the strong convexity parameter of the student loss (if applicable). Under this condition, the tracking error $\|\theta_t - \theta^*(S_t)\|$ remains $O(\sqrt{\alpha_t/\nu_S\mu})$ in expectation.

8. ODE Method

Definition 5.5 (Mean ODE). Consider the continuous-time interpolation of the student update:

$$\frac{d}{d\tau}\theta(\tau) = -\nabla\bar{L}(\theta(\tau))$$

where $\bar{L}(\theta) = \mathbb{E}_{(x,y)\sim P_\infty}[\ell(f_\theta(x), y)]$ is the limiting expected loss, and τ is the continuous-time index scaled by the learning rate.

Theorem 5.4 (ODE Approximation). Under assumptions SE-A5 through SE-A8, and assuming $P_t \rightarrow P_\infty$ sufficiently fast ($\sum \alpha_t \cdot \text{TV}(P_t, P_\infty) < \infty$), the interpolated process $\theta^\alpha(t) = \theta_{\lfloor t/\alpha \rfloor}$ converges weakly to the solution of the ODE:

$$\frac{d}{dt}\theta(t) = -\nabla\bar{L}(\theta(t))$$

as $\alpha \rightarrow 0$, where α is a representative learning rate.

More precisely, define the continuous-time interpolation on $[0, T]$:

$$\theta^\alpha(t) = \theta_k + \frac{t - \tau_k}{\tau_{k+1} - \tau_k}(\theta_{k+1} - \theta_k)$$

where $\tau_k = \sum_{i=1}^k \alpha_i$. Then for any $T < \infty$:

$$\lim_{\alpha \rightarrow 0} \mathbb{P} \left(\sup_{0 \leq t \leq T} \|\theta^\alpha(t) - \theta(t)\| > \delta \right) = 0, \quad \forall \delta > 0$$

Proof sketch. This is the classic ODE method for stochastic approximation (Kushner & Clark, 1978; Benveniste et al., 1990; Kushner & Yin, 2003). The proof involves:

1. **Constructing the interpolation:** $\theta^\alpha(t)$ is a piecewise linear interpolation of the discrete iterates.

2. **Writing the interpolated update:**

$$\theta^\alpha(t) = \theta_0 - \int_0^t \nabla \bar{L}(\theta^\alpha(s)) ds + \text{error terms}$$

3. **Bounding the error terms:**

- Martingale noise term: $M^\alpha(t) = \sum_{\tau_k < t} \alpha_k \xi_k$. By the martingale CLT, $M^\alpha(t) \rightarrow 0$ as $\alpha \rightarrow 0$.
- Discretization error: $O(\max_k \alpha_k)$.
- Distribution shift error: $O(\sum \alpha_k \cdot \text{TV}(P_k, P_\infty))$.

4. **Applying the Arzela-Ascoli theorem** to show tightness of $\{\theta^\alpha\}$, then identifying the limit as the ODE solution via the martingale problem.

Corollary 5.4 (Convergence to Stationary Point). If $\bar{L}$ has a unique global minimum θ^* and is coercive ($\bar{L}(\theta) \rightarrow \infty$ as $\|\theta\| \rightarrow \infty$), then $\theta(t) \rightarrow \theta^*$ as $t \rightarrow \infty$, and consequently $\theta_t \rightarrow \theta^*$ almost surely.

Proposition 5.4 (ODE with Slow Distribution Shift). When the gatekeeper is also evolving, the ODE becomes a coupled system:

$$\begin{aligned}\frac{d}{dt}S(t) &= \nu_S \cdot \Phi(S(t), \theta(t)) \\ \frac{d}{dt}\theta(t) &= -\nabla_{\theta} L_{S(t)}(\theta(t))\end{aligned}$$

where $L_S(\theta) = \mathbb{E}_{(x,y) \sim P_S}[\ell(f_{\theta}(x), y)]$ and Φ is the gatekeeper update vector field.

Interpretation. The full system is a **two-timescale ODE** where the θ dynamics are fast and the S dynamics are slow. This is the continuous-time analog of Theorem 5.3.

9. Connection to State-Conditioned Risk

Definition 5.6 (State-Conditioned NEP Loss). For a given state $s \in \mathcal{S}$ (the state space from the existing SCX theory), define the state-conditioned NEP student loss:

$$L_s(\theta) = \mathbb{E}_{(x,y) \sim P(\cdot | s)}[\ell(f_{\theta}(x), y)]$$

where $P(\cdot | s)$ is the conditional data distribution in state s .

Proposition 5.5 (Decomposition of Expected Loss). The total expected loss for the NEP student decomposes across states:

$$\bar{L}(\theta) = \sum_{s \in \mathcal{S}} \rho_s \cdot L_s(\theta)$$

where $\rho_s = P(X \in s)$ is the state probability. Moreover, the gradient decomposes accordingly:

$$\nabla \bar{L}(\theta) = \sum_{s \in \mathcal{S}} \rho_s \cdot \nabla L_s(\theta)$$

Proof. By the law of total expectation:

$$\bar{L}(\theta) = \mathbb{E}[\ell(f_\theta(X), Y)] = \mathbb{E}[\mathbb{E}[\ell(f_\theta(X), Y) \mid X \in s]] = \sum_s \rho_s L_s(\theta)$$

Gradient commutes with expectation (under Leibniz integral rule, justified by bounded gradient assumption).

Connection to Expert Risk $R_m(s)$. The existing SCX theory defines $R_m(s) = \mathbb{E}_{x \sim P(\cdot|s)}[\ell(f_m(x), f^*(x))]$ as the state-conditioned expert risk. For the NEP student, we analogously define:

$$R_\theta(s) = \mathbb{E}_{(x,y) \sim P(\cdot|s)}[\ell(f_\theta(x), y)]$$

The key difference is that $R_m(s)$ uses the true oracle f^* , while $R_\theta(s)$ uses the observed labels y (which include the gatekeeper's accepted labels). When the gatekeeper is reliable:

$$R_\theta(s) \approx \mathbb{E}_{x \sim P(\cdot|s)}[\ell(f_\theta(x), f^*(x))]$$

i.e., the observed label y is close to the oracle $f^*(x)$.

Theorem 5.5 (Risk Consistency in Self-Evolution). Under the convergence regime $S_t \rightarrow S_\infty$ and $\theta_t \rightarrow \theta^*$, the NEP student’s state-conditioned risk relates to the expert risks as:

$$\lim_{t \rightarrow \infty} R_{\theta_t}(s) \leq \min_m R_m(s) + \varepsilon$$

where $\varepsilon = O(\|\theta^* - \theta_{\text{expert}}\|)$ quantifies the representational gap between the NEP architecture and the best expert.

Proof sketch (conjectured). If the NEP student f_θ has sufficient capacity to represent the best expert f_m , then $\min_\theta R_\theta(s) = \min_m R_m(s)$. The self-evolution process searches over θ , so θ^* should achieve at least the performance of the best expert in each state, provided the memory bank contains sufficient samples from each state. The ε term accounts for approximation error (capacity gap).

10. Full Proof Sketch: Convergence Under Robbins-Monro Conditions

Here we provide a detailed proof sketch for Theorem 5.1.

Setup. Consider the stochastic recursion:

$$\theta_{t+1} = \theta_t - \alpha_t \nabla_\theta \ell(f_{\theta_t}(x_t), y_t)$$

We analyze convergence under RM conditions.

Step 1: Supermartingale construction. Define $V_t = \|\theta_t - \theta^*\|^2$ where θ^* is a stationary point of $\bar{L}$. Compute:

$$\begin{aligned} V_{t+1} &= \|\theta_t - \alpha_t g_t(\theta_t) - \theta^*\|^2 \\ &= V_t - 2\alpha_t \langle \theta_t - \theta^*, g_t(\theta_t) \rangle + \alpha_t^2 \|g_t(\theta_t)\|^2 \end{aligned}$$

Take conditional expectation:

$$\mathbb{E}[V_{t+1} \mid \mathcal{F}_{t-1}] = V_t - 2\alpha_t \langle \theta_t - \theta^*, \nabla L_t(\theta_t) \rangle + \alpha_t^2 \mathbb{E}[\|g_t(\theta_t)\|^2 \mid \mathcal{F}_{t-1}]$$

since $\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = 0$.

Step 2: Gradient inner product bound. By the Lipschitz gradient property:

$$\langle \theta_t - \theta^*, \nabla L_t(\theta_t) \rangle \geq \frac{1}{L_g} \|\nabla L_t(\theta_t)\|^2 - \frac{L_g}{2} \|\theta_t - \theta^*\|^2$$

For the strongly convex case, we use:

$$\langle \theta_t - \theta^*, \nabla L_t(\theta_t) \rangle \geq \mu \|\theta_t - \theta^*\|^2$$

Step 3: Almost sure convergence. Using the supermartingale convergence theorem (Robbins & Siegmund, 1971):

Let $Y_t = V_t$, $X_t = \alpha_t \|\nabla L_t(\theta_t)\|^2$, $Z_t = \alpha_t^2 \|g_t\|^2$.

If: 1. $\mathbb{E}[Y_{t+1} \mid \mathcal{F}_{t-1}] \leq Y_t - X_t + Z_t$ 2. $\sum Z_t < \infty$ almost surely (from $\sum \alpha_t^2 < \infty$ and bounded gradient) 3. $Y_t \geq 0$

Then Y_t converges almost surely and $\sum X_t < \infty$.

Since $\sum \alpha_t = \infty$ and $\sum \alpha_t \|\nabla L_t(\theta_t)\|^2 < \infty$, we must have $\|\nabla L_t(\theta_t)\| \rightarrow 0$ almost surely.

Step 4: Convergence of L_t to $\bar{L}$. From Theorem 5.2:

$$\|\nabla L_t(\theta_t) - \nabla \bar{L}(\theta_t)\| \leq L_g \cdot \text{TV}(P_t, P_\infty) \rightarrow 0$$

Therefore $\|\nabla \bar{L}(\theta_t)\| \rightarrow 0$ almost surely.

Step 5: Rate under strong convexity. If $\bar{L}$ is μ -strongly convex:

$$\|\theta_t - \theta^*\| \leq \frac{2}{\mu} \|\nabla \bar{L}(\theta_t)\|$$

Combining with the gradient convergence rate:

$$\|\nabla \bar{L}(\theta_t)\| = O\left(\frac{1}{\sqrt{t}}\right) \quad \text{in probability}$$

from standard RM analysis, giving $\|\theta_t - \theta^*\| = O(1/\sqrt{t})$.

11. What Breaks When Distribution Shifts Too Rapidly

Definition 5.7 (Critical Drift Rate). Define the data distribution drift per student step:

$$\delta_t = \text{TV}(P_{t+1}, P_t)$$

The drift is **slow** if $\delta_t \ll \alpha_t$ (the distribution changes slower than the gradient step). The drift is **rapid** if $\delta_t \gtrsim \alpha_t$ or larger.

Theorem 5.6 (Failure Under Rapid Shift). If the data distribution shifts too rapidly relative to the learning rate:

$$\limsup_{t \rightarrow \infty} \frac{\delta_t}{\alpha_t} = \infty$$

then the student may fail to converge. Specifically:

(a) **Oscillation.** The parameter sequence may enter a limit cycle:

$$\liminf_{t \rightarrow \infty} \|\theta_t - \theta^*(P_t)\| > 0$$

(b) **Divergence.** In the worst case, the loss may diverge:

$$\bar{L}(\theta_t) \rightarrow \infty \quad \text{as } t \rightarrow \infty$$

(c) **Catastrophic forgetting.** Earlier-learned states may be overwritten as the distribution shifts toward new states:

$$L_s(\theta_t) \rightarrow \infty \quad \text{for states } s \text{ that disappear from } P_t$$

Proof sketch. The ODE approximation (Theorem 5.4) requires that the drift δ_t is summable against α_t . When $\delta_t \gg \alpha_t$, the ODE's vector field changes faster than the gradient flow can follow, violating the conditions for weak convergence. This is analogous to the failure of stochastic gradient descent under non-stationary distributions when the learning rate is too small to track the drift (Kushner & Yin, 2003, Chapter 8; Borkar, 2008, Chapter 6).

Practical Implications for SCX Self-Evolution:

1. **Gatekeeper oscillations:** If S_t oscillates (regime 2 in Theorem SE-1), the distribution P_t oscillates, preventing student convergence.
2. **Perpetual discovery:** If $\mathcal{M}_t$ grows by adding qualitatively new types of data (regime 3), the distribution is genuinely non-stationary, and the student must use a constant (non-decaying) learning rate.
3. **Divergent collapse** (regime 4): If S_t becomes degenerate (rejecting all data), the distribution collapses and the student receives no training signal.

Proposition 5.6 (Adaptive Learning Rate Heuristic). To handle moderate drift, the learning rate should be:

$$\alpha_t = \min \left(\alpha_0, \frac{C}{\sqrt{\text{Var}_{P_t}[\nabla \ell]}} \right)$$

where C is a constant and Var_{P_t} is the gradient variance under the current distribution. This is the **AdaGrad**-style heuristic, which reduces the learning rate when gradients are consistent (convergence) and increases it when the distribution shifts (drift).

Status. This heuristic is well-motivated but lacks a formal convergence guarantee in the self-evolution context.

12. Summary of Proven vs. Conjectured Claims

Claim	Status	Notes
Theorem 5.1: RM convergence (stationary)	Proven	Standard Robbins-Monro theory; requires stationary P_∞
Theorem 5.1: Gradient vanishing	Proven	Supermartingale convergence argument
Theorem 5.1: Rate under strong convexity	Proven	Standard RM rate $O(1/\sqrt{t})$
Theorem 5.2: Distribution shift bound	Proven	From TV convergence of P_t to P_∞
Theorem 5.3: Two-timescale convergence	Proven under separation condition	Singular perturbation + Tikhonov theorem
Theorem 5.4: ODE approximation	Proven	Kushner & Yin (2003) framework; requires drift summability
Proposition 5.3: Cumulative shift error	Proven	Direct bound from Theorem 5.2
Theorem 5.5: Risk consistency	Conjectured	Requires capacity + representability assumptions
Theorem 5.6: Failure under rapid shift	Proven	Standard non-stationary SA theory

Claim	Status	Notes
Proposition 5.1: Constant vs. decaying LR	Proven	Classic result from SA theory
Proposition 5.2: Gradient consequences	Proven	Direct from Lipschitz assumption
Proposition 5.4: Coupled ODE	Proven	Continuous-time limit of two-timescale system
Proposition 5.6: Adaptive learning rate	Heuristic	Lacks formal convergence guarantee

Key uncertainty: The risk consistency (Theorem 5.5) requires the NEP student to have sufficient capacity to match the oracle expert, which depends on the specific architecture. The adaptive learning rate heuristic (Proposition 5.6) is practically motivated but unproven.

13. References

1. Robbins, H., & Monro, S. (1951). A stochastic approximation method. *Annals of Mathematical Statistics*, 22(3), 400-407.
2. Robbins, H., & Siegmund, D. (1971). A convergence theorem for non-negative almost supermartingales. *Optimization Methods in Statistics*, 233-257.
3. Kushner, H. J., & Clark, D. S. (1978). *Stochastic Approximation Methods for Constrained and Unconstrained Systems*. Springer.

4. Kushner, H. J., & Yin, G. G. (2003). *Stochastic Approximation and Recursive Algorithms and Applications* (2nd ed.). Springer.
5. Benveniste, A., Metivier, M., & Priouret, P. (1990). *Adaptive Algorithms and Stochastic Approximations*. Springer.
6. Borkar, V. S. (2008). *Stochastic Approximation: A Dynamical Systems Viewpoint*. Cambridge University Press.
7. Khalil, H. K. (2002). *Nonlinear Systems* (3rd ed.). Prentice Hall.
8. Bottou, L. (1998). Online learning and stochastic approximations. *Online Learning in Neural Networks*, 9-42.
9. Bottou, L., & LeCun, Y. (2005). On the convergence of online gradient descent for neural networks. *Neurocomputing*, 63, 295-316.
10. Duchi, J., Hazan, E., & Singer, Y. (2011). Adaptive subgradient methods for online learning and stochastic optimization. *JMLR*, 12, 2121-2159.
11. Kingma, D. P., & Ba, J. (2015). Adam: A method for stochastic optimization. *ICLR 2015*.
12. Polyak, B. T., & Juditsky, A. B. (1992). Acceleration of stochastic approximation by averaging. *SIAM Journal on Control and Optimization*, 30(4), 838-855.
13. Yin, G. G., & Zhu, C. (2010). *Hybrid Switching Diffusions: Properties and Applications*. Springer.

End of 05_stochastic_approximation.md – Stochastic approximation analysis of NEP student evolution.